

Code-Layout, Readability and Reusability

Date	29 June 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	yes	Python code follows standard 4-space indentation.
2	Proper File Structure	Files and folders are logically organized	yes	Project organized into Flask, Dataset, Training, Templates, Static, and Documentation folders.
3	Meaningful Variable Names	Variables reflect their purpose clearly	yes	Variables such as model, prediction, country, and hdi_score are descriptive.
4	Function / Method Names	Functions are descriptively named	yes	Functions like predict() and route handlers clearly indicate their purpose.
5	Code Comments	Inline and block comments explain logic	Partial	Important sections are commented; additional comments can improve readability.
6	Modular Design	Code is split into reusable functions/modules	yes	Model training, Flask application, and templates are maintained separately.
7	No Redundant Code	Duplicate or unused code is removed	yes	Only necessary code is retained in the project.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic input validation is implemented; more exception handling can be added.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Linear Regression Model	Python, Scikit-learn	Prediction Module	High

2	Flask Application	Python, Flask	Web Interface	High
3	Data Preprocessing Script	Python, Pandas	Model Training	High
4	HTML Templates	HTML, CSS, Bootstrap	User Interface	High
5	Pickle Model (model.pkl)	Pickle	Prediction Engine	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate modules.
Readability	5	Clear naming conventions and easy-to-understand code.
Reusability	5	Reusable model, templates, and preprocessing modules.
Documentation / Comments	4	Adequate documentation; additional comments could improve maintainability.
Overall Score	4.8	The codebase is well-structured, readable, reusable, and suitable for future enhancements.